

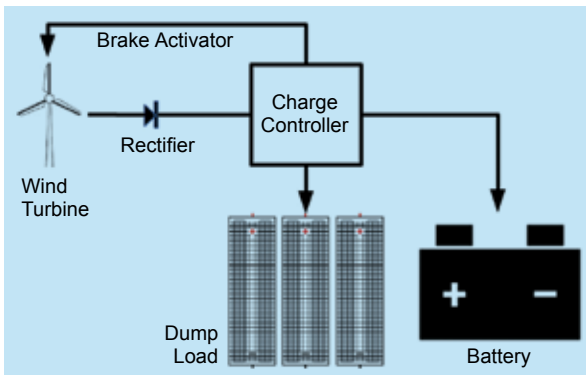


Fig. 2: Windmill charge controller with dummy load

the pen is lightweight so I can carry it or write with it easily. It has a nice grip that is comfortable to hold for hours. It writes smoothly. The cap is easy to open but at the same time remains secure once closed. The clip, which is visible from my pocket, is aesthetically designed and acts as a fashion accessory.

The ability to use a product in different ways increases its utility and makes us use them frequently. Which in turn makes them more adorable. Now, the product does not become useful by the mere presence of the affordance. The user of the product must know about its presence and get a hint on how to use the affordance.

The feature that communicates affordance to the user is called a *signifier*. For instance, in my pen, there is a ring between the body and the cap. This is a signifier that separates two similar-looking objects so that I know which two parts to hold and pull to open the cap. Affordances are the physical features that enable us to use the product; signifiers are the communication features of the affordances that tell us about the presence of the feature and give hint on how to use the affordances.

Proper design of signifiers differentiates great products from the rest. Improper signifiers, on the other hand, create difficulty in using the product and reduce its value. For

A pedal gives a clear indication of pressing it with the foot.

It is not uncommon to find improper use of signifiers. When you see a door with a bar your instinct will be to push it. But if you have to pull it to open you get into a behavioural conflict. Faced with a conflict of the signifiers, our first reaction will be confusion. If we have to use such features frequently then instead of loving we shall develop a hatred for the product.

We are all too familiar with the elaborate ornamentation around the entrances of a building. The ornamentations help to separate the gate from the fence, or the door from the wall. While doors and gates are the affordances, ornamentation that is around the door or gate is the signifier. Just like the ornamentation and design of the doors and gates give us an indication of the ambiance and quality of what is inside the house, the signifiers in the product give us an indication of the product.

Aesthetics and quality of interface play a very important role to make a product successful. Before we discuss product design let us first discuss the concept of a product.

A product is meant to achieve certain results. The product can be a physical thing or it can be a software. Television, computer, mobile phone, fridge, and cars are examples of physical products. Operating systems, computer games, and

instance, when we see a door recessed inside a frame, the signifier invites us to push it open. If the door opens outside, or we need to pull it sideways, then that needs to have a certain signifier telling us that. A bar on the door signifies pushing and a handle invites us to pull it.

mobile applications are examples of software products.

We use a product to achieve something. This results in the utility of the product. We value the product because of the utility that it has. Now, there are certain costs that we have to incur to use the product. These costs are in the form of money, energy, time, and attention that we have to spend to get our desired result. We like the product that gives more utility using less cost.

A lovable product has to provide more utility at a lesser cost, which gives us the perceived utility. It also needs to get used frequently to be in the habit-forming zone. Such a combination needs careful planning and careful trade-offs between various alternatives. Swiss knife or combination tools are one of the best examples of clever design to increase the frequency of use.

In my desk's drawer, I have a supply of scissors, a knife, a magnifying glass, and other such tit-bit tools. There is also a Swiss knife that has all the functions. It is no surprise that most of the time I will grab the Swiss knife for anything. As it is frequently used, it is kept in the most accessible part of my drawer. I can grab the Swiss knife even without looking at it. Any of the other tools will require a search inside my drawer.

If you look at a catalogue of Swiss knives, you will find there are plenty of different varieties of the knife. Within the knife the blades and implements provided in different knives are carefully planned. For instance, the one I have is meant for an urban office and has a reading glass, a few screwdrivers, a knife, a can opener, and other things.

I have another version of the knife, which is meant for camping. That one comes with a tiny saw, a file, and a hook, which are missing from the office version. Each of these has the appropriate set of tools